

Homework 1

PSTAT 131/231

Machine Learning Main Ideas

Please answer the following questions. Be sure that your solutions are clearly marked and that your document is neatly formatted.

You don't have to rephrase everything in your own words, but if you quote directly, you should cite whatever materials you use (this can be as simple as a note showing from which lecture or page # of the book you use).

Question 1:

Define supervised and unsupervised learning. What are the difference(s) between them?

Type your answer here

Supervised Learning: Supervised learning is basically when you train a model using labeled data (each input comes with a known output/target). -The model learns a mapping: $X \rightarrow Y$ - The goal is to predict the correct output for new, unseen data -Some examples could be predicting home prices or classifying emails as spam/not-spam

Unsupervised Learning: Unsupervised learning uses unlabelled data (meaning there is no predetermined output).

- The model tries to discover patterns/structures/relationships that can be in the data
- The goal is to discover hidden features/groupings
- Some examples could be finding patterns in user behavior

Question 2:

Explain the difference between a regression model and a classification model, specifically in the context of machine learning.

Type your answer here

Regression Model: A regression model is used when the output (target variable) is continuous (numerical).

- It predicts actual values along a range
- Output can be any real #

- Some examples of a regression model can be: predicting a baseball player's batting average or forecasting temperature

Classification Model: A classification model is used when the output is categorical (labels/classes)

- It predicts which category an observation belongs to
- Output will be a finite set
- An example could be to classify if an email is spam vs not spam

Question 3:

Name two commonly used metrics for regression ML problems. Name two commonly used metrics for classification ML problems.

Common Metrics:

Regression: Mean Square Error (MSE), Root Mean Squared Error (RMSE) Classification: Accuracy, AOC-ROC (Area Under the ROC Curve)

Question 4:

As discussed, statistical models can be used for different purposes. These purposes can generally be classified into the following three categories. Provide a brief description of each.

Descriptive models: Descriptive models are used to summarize or visualize patterns from the main characteristics of a dataset. The goal is not prediction but to highlight trends or relationships. -Example: Using a histogram to show the distribution of heights in a population

Inferential models: Inferential models are used to draw conclusions about a larger population based on a smaller sample of data. This model type is used to test hypotheses and understand relationships between different variables (p-values, confidence intervals, testing) -Example: Conducting a t-test to determine if a new medication is significantly more effective than a placebo

Predictive models: Predictive models focus on forecasting future outcomes or unobserved values. These models emphasize on minimizing prediction error. -Example: Using a Random Forest Classifier to estimate the probability that a prospective athlete will succeed at the professional level based on their historical performance metrics.

Question 5:

Predictive models are frequently used in machine learning, and they can usually be described as either mechanistic or empirically-driven. Answer the following questions.

Define mechanistic. Define empirically-driven. How do these model types differ? How are they similar?

Mechanistic (parametric) models: assume a specific functional form (linear relationship) because you believe that is how the underlying system actually works. These models are built on pre-existing theory.

Empirically-Driven Models: these models don't start with theory. Instead, empirically data-driven models use algorithms to "discover" patterns and function forms directly from the data. They are also flexible and can adapt to complex non-linear shapes.

These models are similar because they both require data for validation and aim to minimize the difference between predicted/actual values. Also both use optimization techniques to find the best parameters.

In general, is a mechanistic or empirically-driven model easier to understand? Explain your choice.

In my opinion I believe mechanistic models are significantly easier to understand. This is because they rely on specific mathematical equations. For example in linear regression the slope will tell you exactly how much the output changes for every unit of input x . Empirically driven models may involve thousands of hidden interactions that make it super difficult to understand.

Describe how the bias-variance tradeoff is related to the use of mechanistic or empirically-driven models.

Mechanistic Models → Higher bias, lower variance.

- Might miss true relationships if the real world doesn't follow theory.
- Low variance because model won't change drastically if you train on different set of data

Empirically-driven → Lower bias, higher variance.

- Low bias because they are flexible to fit almost every data shape
- High variance (overfitting). Can memorize the noise in your specific data causing the model to perform poorly when applied to unseen data.

Question 6:

A political candidates campaign has collected some detailed voter history data from their constituents. The campaign is interested in two the two questions below. Classify each question as either predictive or inferential. Explain your reasoning for each.

Given a voters profile/data, how likely is it that they will vote in favor of the candidate?

Predictive

The goal is to use existing voter data to produce an accurate probability estimate of a future outcome (voting behavior). There's no interest in why or which variables matter.

How would a voters likelihood of support for the candidate change if they had personal contact with the candidate?

Inferential

This question asks about the effect of a specific variable (personal contact) on the outcome (likelihood of support). The goal is to understand the relationship and potentially establish causality (does personal contact actually change support?). This requires hypothesis testing and understanding the statistical significance of that relationship.

Exploratory Data Analysis

This section will ask you to complete several exercises. For this homework assignment, we'll be working with the mpg data set that is located in the data folder within the homework assignment. To load, manage and visualize our data in we will be using the pandas, matplotlib and seaborn libraries.

Exploratory data analysis (or EDA) is not based on a specific set of rules or formulas. It is more of a state of curiosity about data. It's an iterative process of:

- generating questions about data
- visualizing and transforming your data as necessary to get answers
- using what you learned to generate more questions

A couple questions are always useful when you start out. These can include questions about what variation and covariation occurs between the variables.

```
In [3]: # Initial imports
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns
```

Loading in the data

Pandas as a popular library for manipulating and managing data in Python. Throughout this course you will get more acquainted with it and its uses. Please make sure to run the code

chunk above, to access pandas and matplotlib functions we will use the alias' pd and plt.

Refer to pandas [read_csv\(\)](#) for documentation on how we can load in our data at

'data/mpg.csv'.

```
In [7]: # Read in the data here
mpg_data = pd.read_csv("/Users/clee/Desktop/PSTAT_131/homework-1-python/data/mpg.csv")
# Calling the head in a Jupyter notebook will give us a brief look at our data
mpg_data.head()
```

```
Out[7]:
```

	manufacturer	model	displ	year	cyl	trans	drv	cty	hwy	fl	class
0	audi	a4	1.8	1999	4	auto(l5)	f	18	29	p	compact
1	audi	a4	1.8	1999	4	manual(m5)	f	21	29	p	compact
2	audi	a4	2.0	2008	4	manual(m6)	f	20	31	p	compact
3	audi	a4	2.0	2008	4	auto(av)	f	21	30	p	compact
4	audi	a4	2.8	1999	6	auto(l5)	f	16	26	p	compact

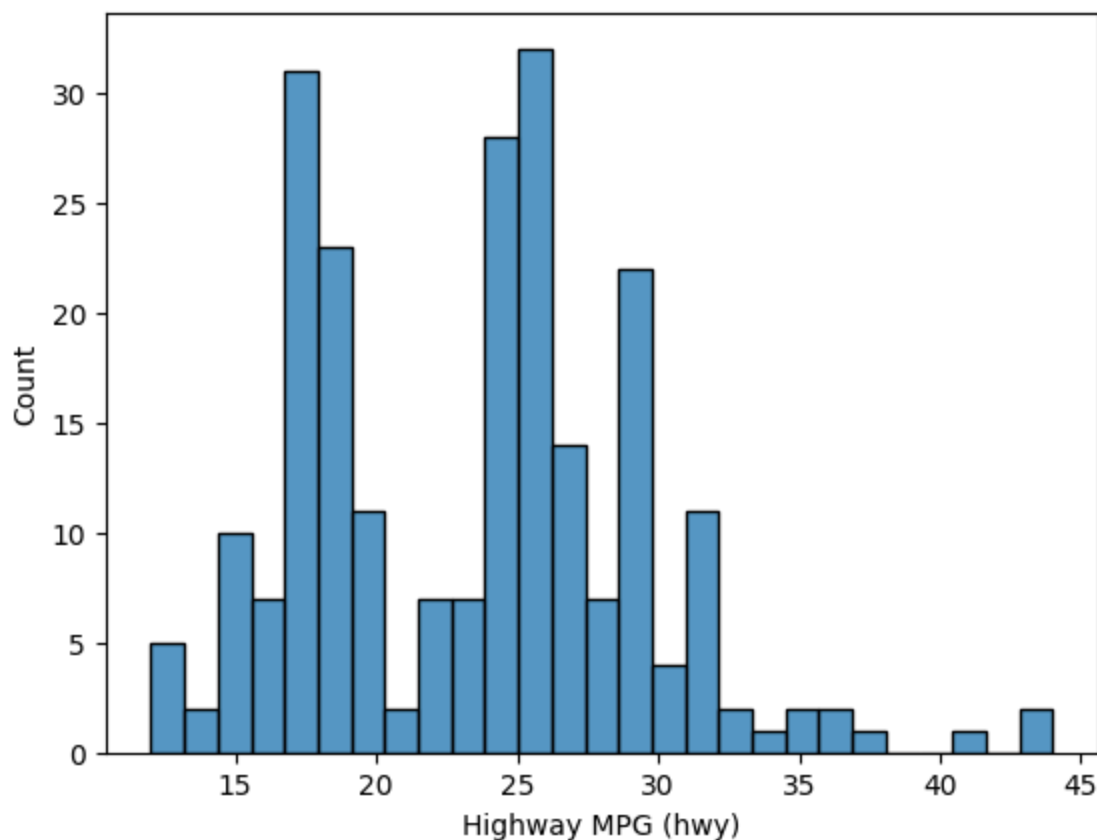
Question 7:

We are interested in highway miles per gallon, or the hwy variable. Create a histogram of this variable. Describe what you see/learn. Refer to [sns.histplot\(\)](#), [plt.xlabel](#) and [plt.ylabel](#) for documentation on creating histograms with matplotlib.

Hints:

- To access a column of a dataframe use `dataframe['column']`
- Consider setting the bins parameter to the number of unique values in `hwy`
- Read what the `.unique()` function does

```
In [10]: # Create a histogram
sns.histplot(x=mpg_data['hwy'], bins=len(mpg_data['hwy'].unique()))
# Fill in labels
plt.xlabel('Highway MPG (hwy)')
plt.ylabel('Count');
```



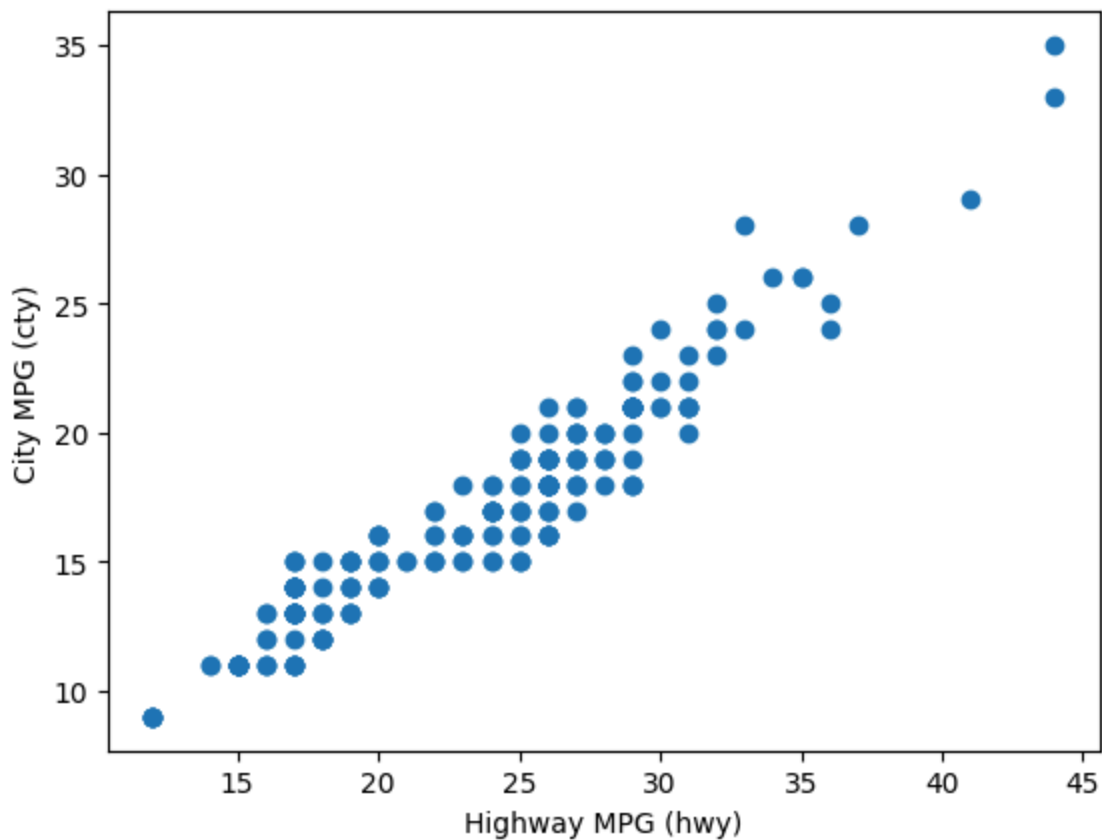
The distribution of highway MPG is roughly right-skewed. Most vehicles achieve between 15–30 highway MPG, with the most common values clustering around 26–28 MPG. Very few vehicles fall below 15 or above 40 MPG. Overall, the majority of vehicles in this dataset have moderate fuel efficiency.

Question 8:

Create a scatterplot. Put hwy on the x-axis and cty on the y-axis and use axis labels. Describe what you notice. Is there a relationship between hwy and cty? What does this mean?

Refer to [plt.scatter\(\)](#) for documentation on creating scatterplots with matplotlib.

```
In [12]: # Create the scatterplot and add labels
plt.scatter(x=mpg_data['hwy'], y=mpg_data['cty'])
plt.xlabel('Highway MPG (hwy)')
plt.ylabel('City MPG (cty)');
```



There is a clear strong positive linear relationship between highway MPG (hwy) and city MPG (cty). As highway fuel efficiency increases, city fuel efficiency also increases. This makes sense because vehicles that are more fuel efficient on the highway tend to also be more fuel efficient in the city,

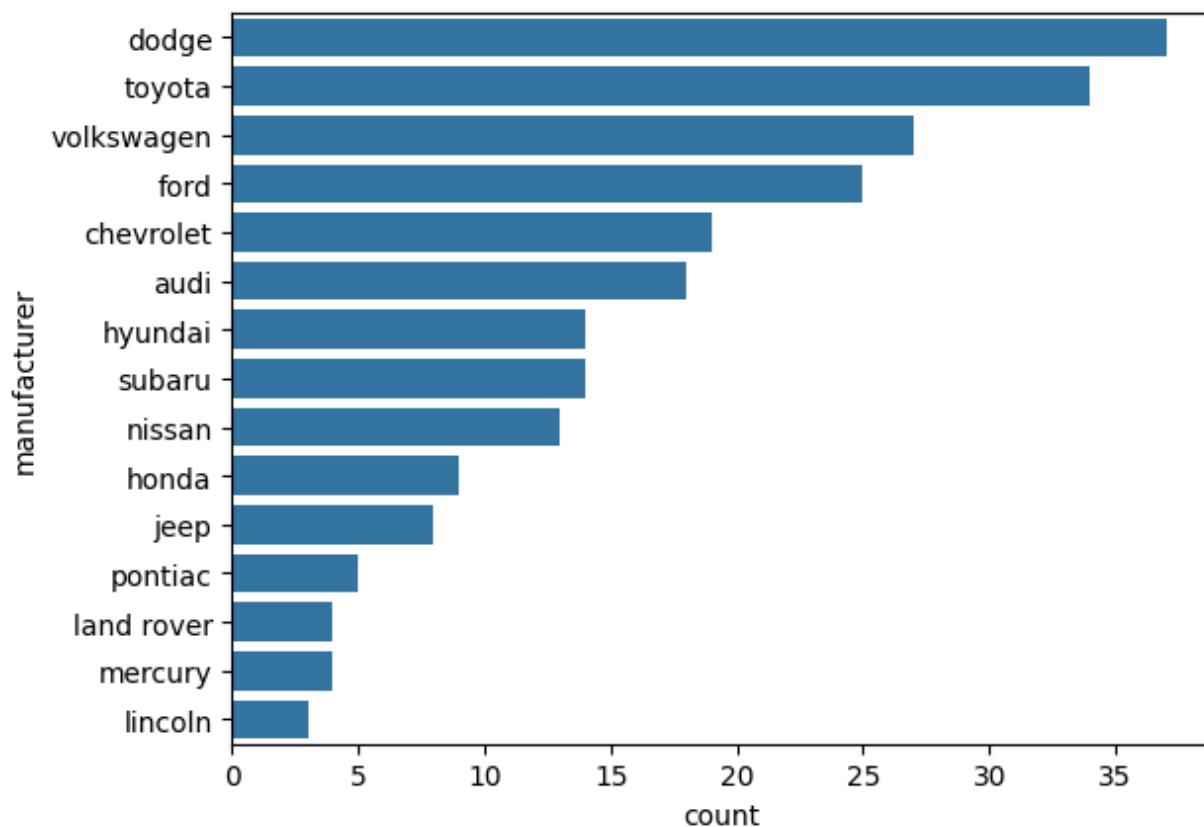
Question 9:

Use `sns.countplot` to make a bar plot of manufacturer. Flip it so that the manufacturers are on the y-axis. Order the bars by height. Which manufacturer produced the most cars? Which produced the least?

Refer to [sns.countplot\(\)](#) for documentation on creating boxplots with seaborn.

Refer to [.value_counts\(\)](#) and [.index](#) for documentation on how we can add an ordering to our bar plot.

```
In [13]: sns.countplot(y=mpg_data['manufacturer'], order=mpg_data['manufacturer'].value_
Out[13]: <Axes: xlabel='count', ylabel='manufacturer'>
```



```
In [14]: mpg_data['manufacturer'].value_counts()
```

```
Out[14]: manufacturer
dodge      37
toyota     34
volkswagen 27
ford       25
chevrolet  19
audi       18
hyundai    14
subaru     14
nissan     13
honda      9
jeep       8
pontiac    5
land rover 4
mercury    4
lincoln    3
Name: count, dtype: int64
```

Dodge produced the most cars in this dataset with 37 vehicles, followed closely by Toyota (34) and Volkswagen (27).

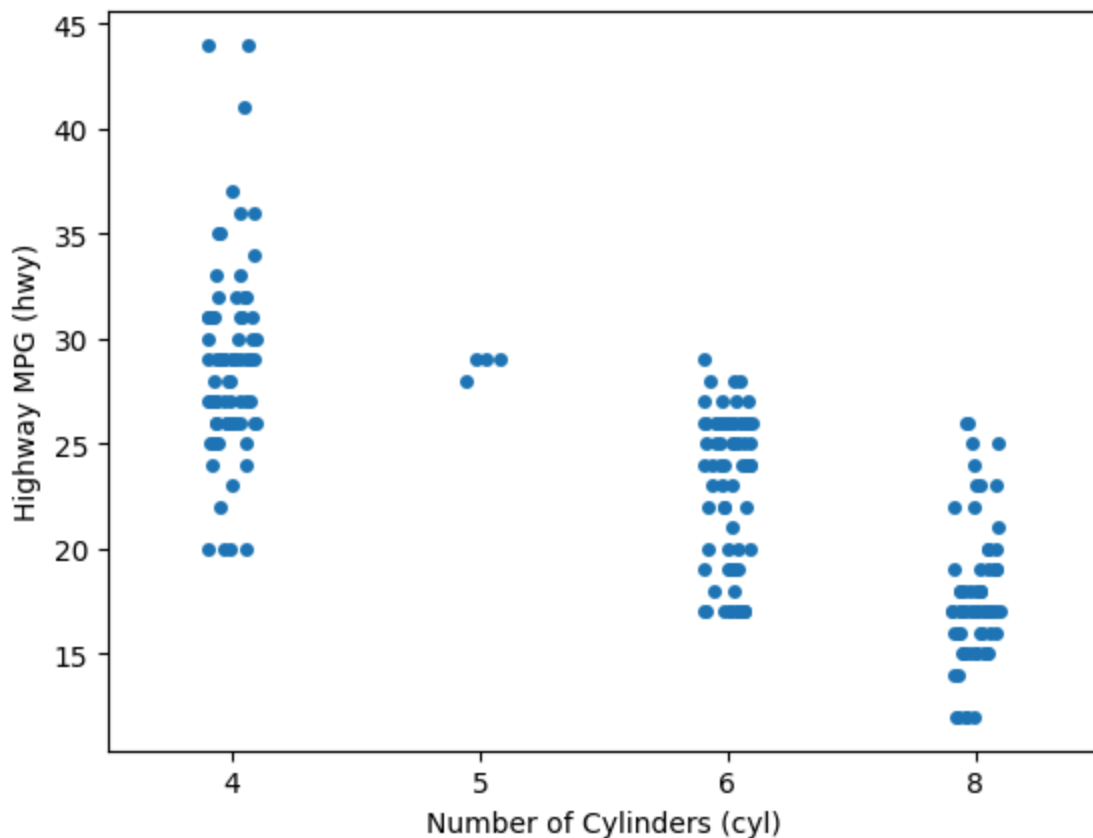
Lincoln produced the fewest cars with only 3 vehicles, followed by Mercury and Land Rover (4 each).

Question 10:

Make a jitter plot of hwy, grouped by cyl and add labels. Describe what you see. Is there a relationship between hwy and cyl? What do you notice?

Refer to [sns.stripplot\(\)](#) for documentation on creating jitterplots with seaborn.

```
In [15]: sns.stripplot(x=mpg_data['cyl'], y=mpg_data['hwy'])  
plt.xlabel('Number of Cylinders (cyl)')  
plt.ylabel('Highway MPG (hwy)');
```



There is a clear negative relationship between number of cylinders (cyl) and highway MPG (hwy). As the number of cylinders increases, highway fuel efficiency decreases.

4-cylinder vehicles have the highest highway MPG (typically 20–35 MPG)

6-cylinder vehicles fall in the middle (typically 17–29 MPG)

8-cylinder vehicles have the lowest highway MPG (typically 12–25 MPG)

Question 11:

Use a seaborn heatmap to make a lower triangle correlation matrix of the mpg dataset with the `annot` parameter set to true.

To assist in making the correlation matrix we will be utilizing numpy which is a library that adds support for large, multi-dimensional arrays and matrices, along with a large collection of high-level mathematical functions. Numpy is loaded with the alias of `np`.

Use `.select_dtypes()` to filter out all non-numeric data in the dataframe.

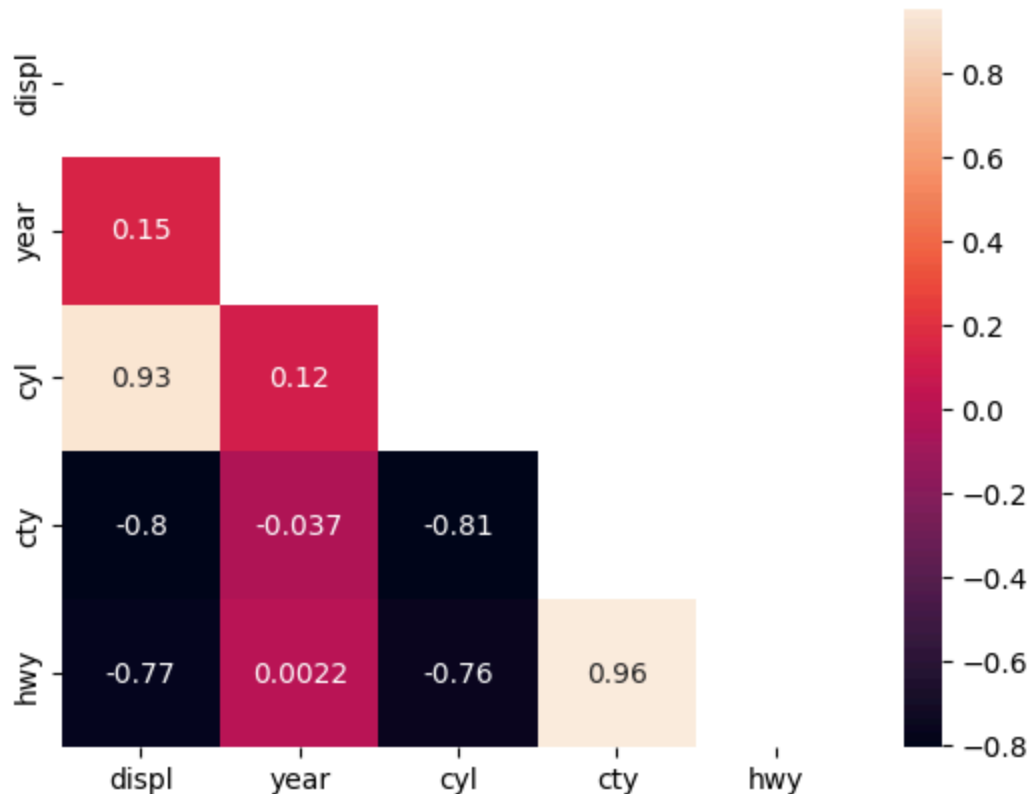
Use the pandas `.corr()` function to get the correlation of a dataframe.

Use numpy's `np.triu()` function to create a triangle mask so that the heatmap is lower triangular.

Refer to `sns.heatmap()` for documentation on how to make a heatmap.

Which variables are positively or negatively correlated with which others? Do these relationships make sense to you? Are there any that surprise you?

```
In [16]: mpg_numeric = mpg_data.select_dtypes(include='number')
mask_matrix = np.triu(np.ones_like(mpg_numeric.corr(), dtype=bool))
sns.heatmap(mpg_numeric.corr(), annot=True, mask=mask_matrix);
```



Positive correlations: -cty and hwy are very strongly positively correlated ($r \approx 0.96$) — as seen in Q8, vehicles efficient on the highway are also efficient in the city -displ and cyl are strongly positively correlated ($r \approx 0.93$) — more cylinders means a larger engine displacement, which makes sense mechanically -year has a slight positive correlation with cty and hwy, suggesting newer vehicles are marginally more fuel efficient

Negative correlations: -ispl is strongly negatively correlated with both cty ($r \approx -0.80$) and hwy ($r \approx -0.77$) — larger engine displacement leads to lower fuel efficiency -cyl is strongly negatively correlated with both cty ($r \approx -0.81$) and hwy ($r \approx -0.76$) — more cylinders leads to lower fuel efficiency

SOURCES: I used Claude (LLM) for coding assitacne and EDA visualizations (Questions 7-14)

231 Students Only:

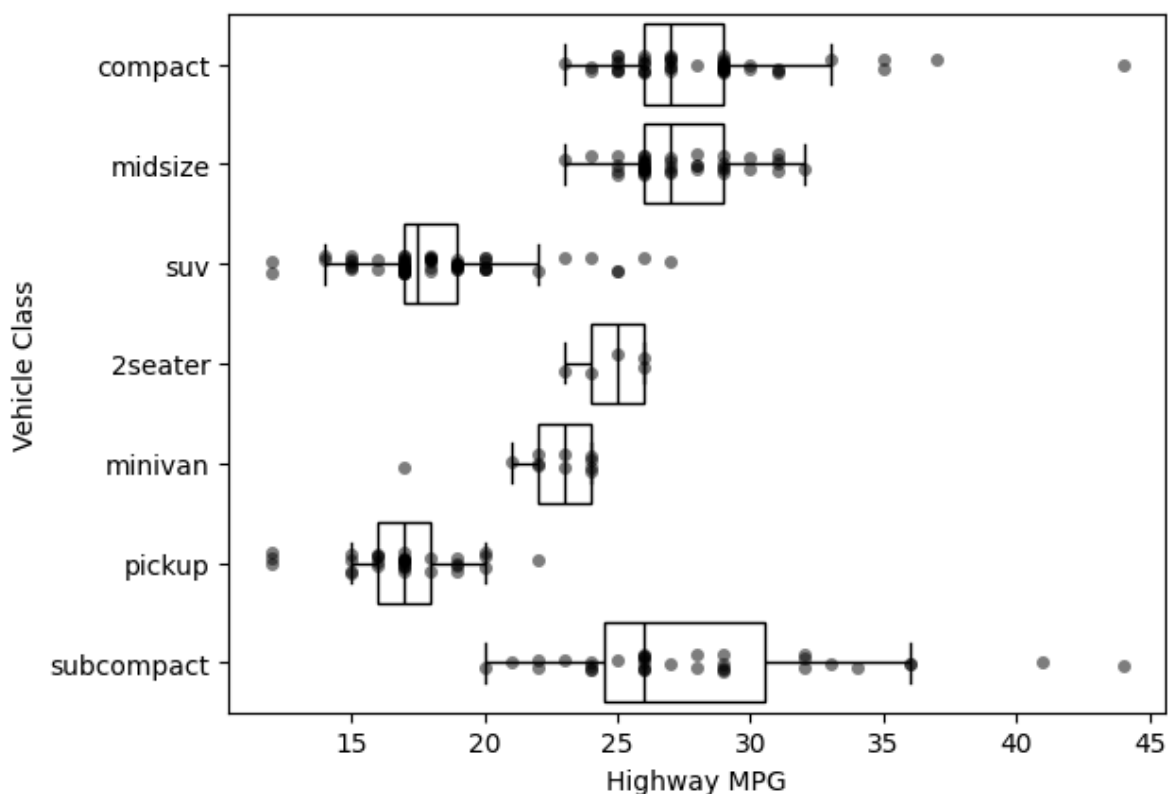
Use seaborn to recreate graphics.

What might be helpful:

- `sns.boxplot()`
- `sns.lineplot()`
- If seaborn graphs are created subsequently they will layer on top of one another.
- Utilize arguments like `linecolor`, `color`, `palette`, `style`, `hue` and `hue_order` when necessary.

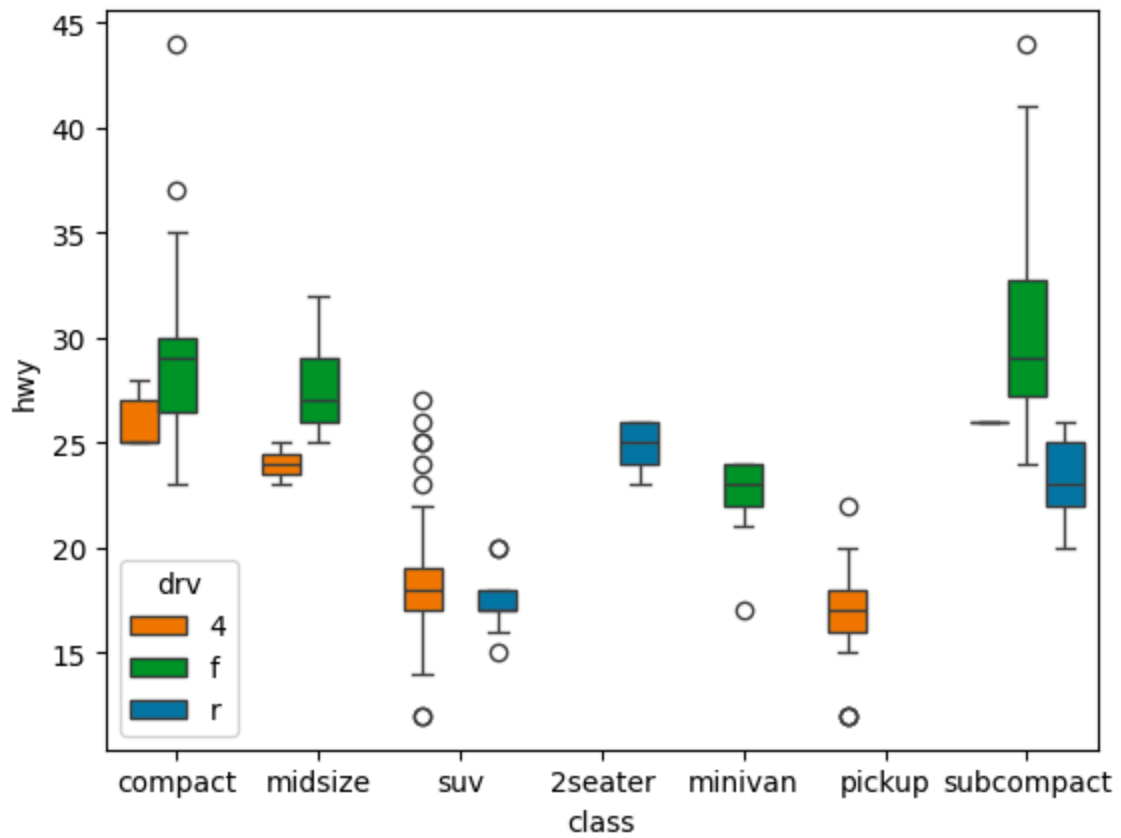
Question 12:

Recreate the following graphic, as closely as you can.



Question 13:

Recreate the following graphic.



Question 14:

Recreate the following graphic.

